

Technical Discussion: Exploratory Data Analysis (EDA) of Acoustic Signatures

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1. Introduction

This document presents a comprehensive technical discussion based on the Exploratory Data Analysis (EDA) phase of the underwater vessel noise classification project. The observations are derived from the spectral and temporal analysis of the ShipsEar corpus.

2. Time-Domain Waveform Characteristics

The analysis of raw acoustic pressure waves reveals distinct amplitude envelopes. As shown in **Figure 1**, the temporal structures vary significantly.

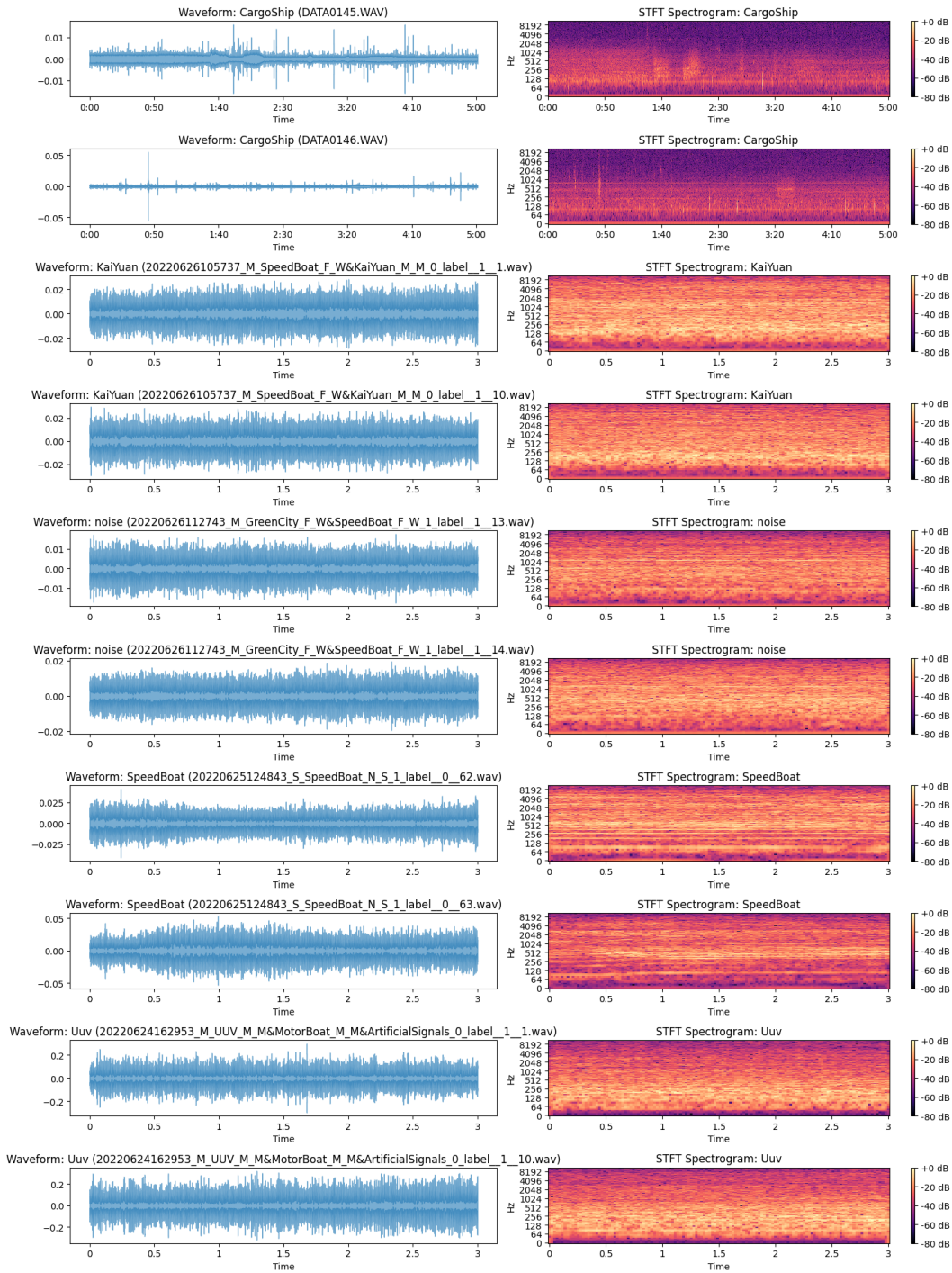


Figure 1: Comparative Analysis of Time-Domain Waveforms and Log-Mel Spectrograms.

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- **CargoShip/KaiYuan:** Continuous, dense envelopes indicating steady-state propulsion.
- **SpeedBoat:** High-variance transient spikes caused by hull-slamming.
- **Uuv:** Stealthy, low-energy signatures with minimal mechanical transients.

3. Spectrogram & SNR Observations

Log-Mel Spectrograms identify anthropogenic tonals. The Average Power Spectral Density (PSD) in **Figure 2** quantifies signal strength.



Figure 2: Average PSD Comparison highlighting SNR profiles.

$$SNR_{dB} f = 10 \log_{10} \left(\frac{P_c(f)}{P_N(f)} \right) \quad (1)$$

4. Statistical Distribution and Class Imbalance

Class Label	Count	Weight
KaiYuan	30	21.4%
Noise	30	21.4%
SpeedBoat	30	21.4%
Uuv	30	21.4%
CargoShip	20	14.3%
Total	140	100.0%

Table 1: Class Distribution Statistics

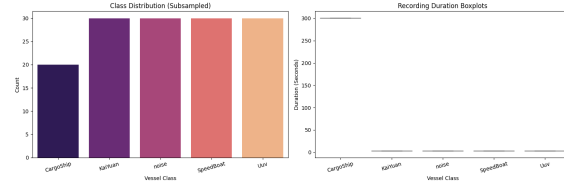


Figure 3: Visualizing Class Representation

5. Directives for Feature Engineering

Key Engineering Takeaways

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1. **Mel-Scale Utilization:** Emphasizes low-frequency anthropogenic tonals.
2. **Dynamic Features:** Delta MFCCs for transient detection.
3. **Evaluation Strategy:** **Macro F1-Score** is the primary metric.